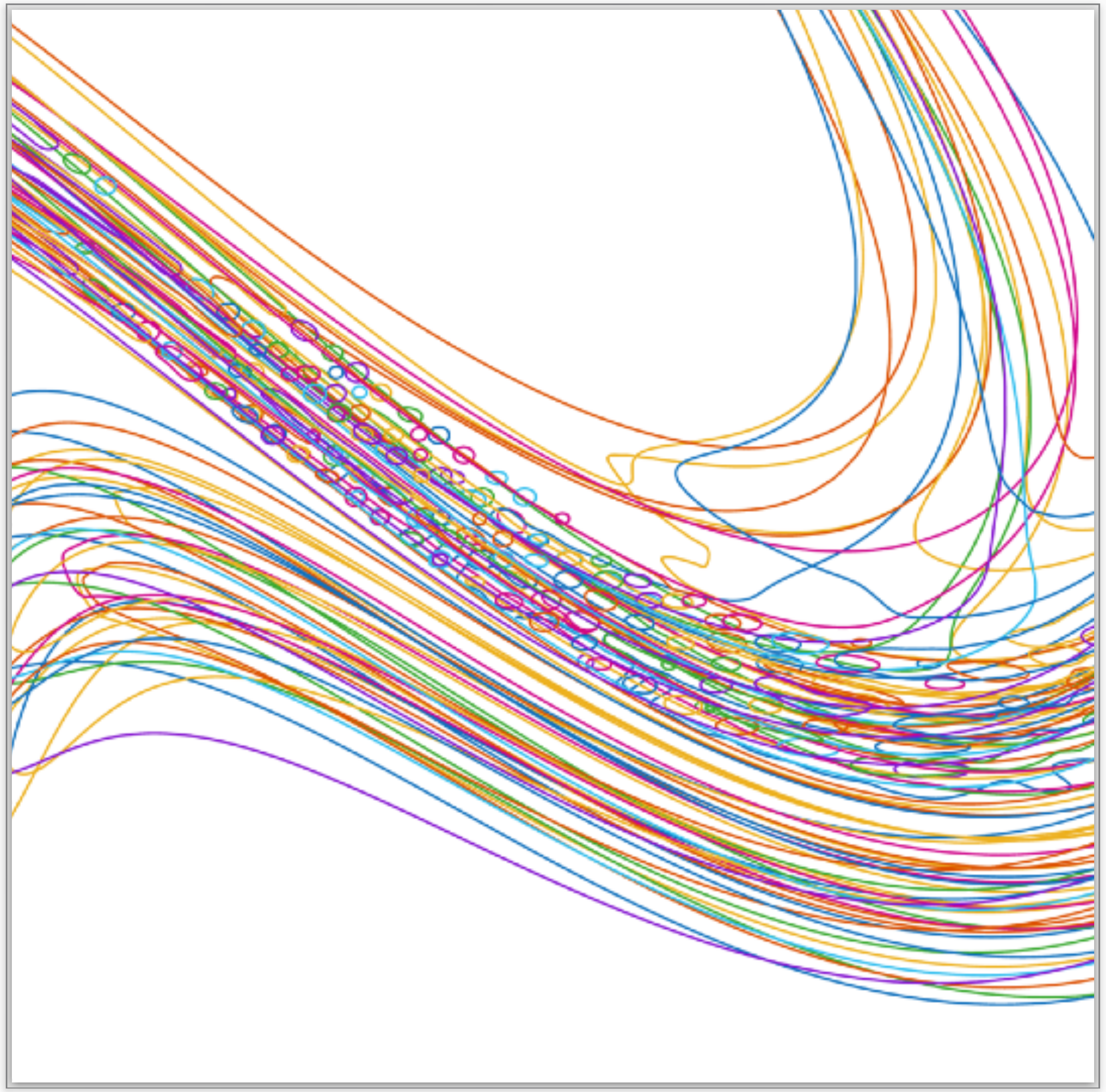
MICROSCALE TOPOLOGICAL SURGERY FOR TRACKING FILAMENT BREAKUP



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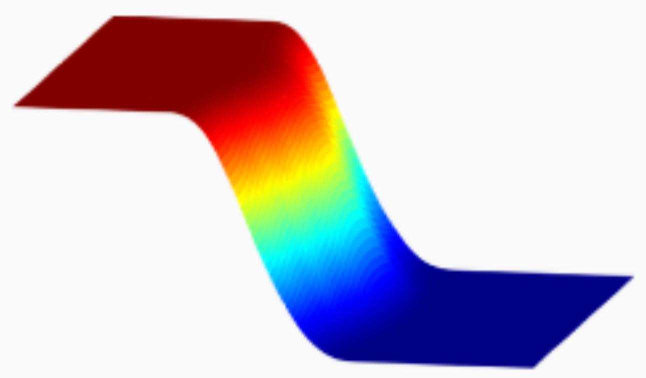
EULERIAN CAPTURING

Interface is implicitly captured by a level-set function on a background mesh



$$\Gamma(t) = \left\{x \in \mathbb{R}^d : \phi(x, t) = 0\right\}$$

$$\partial_t \phi + u \cdot \nabla \phi = 0$$

LAGRANGIAN TRACKING

Interface is explicitly tracked as a codimension—1 parametrized surface

$$\Gamma(t) = \left\{\gamma(s, t) : s \in \mathbb{S}^{d-1}\right\}$$

$$\partial_t \gamma = u \circ \gamma$$